

Precalc Lesson 1.5 Homework

Answer
Key

1. a. $y = |x|$, horizontal shift left 9
b. $y = x^3$, vertical stretch by a factor of 5, vertical shift down 8.

2. $y = 6(x - 4)^3 - 12$ or $y = 6((x - 4)^3 - 2)$

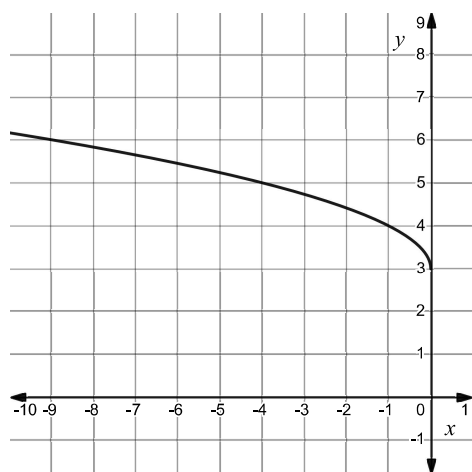
3. B

4. Domain: All real numbers
Range: $y \leq 6$

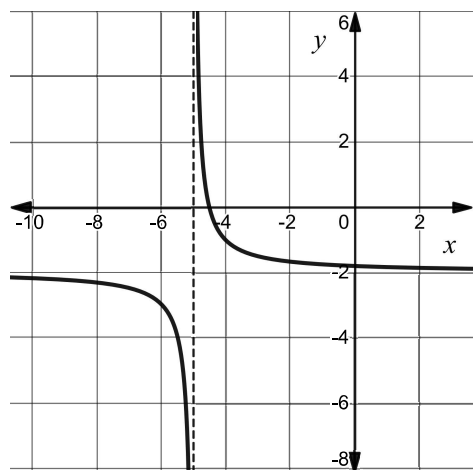
5. Domain: $x \geq -4$
Range: $y \geq 1$

6. The graph of $f(x) = \frac{1}{x-2} + 7$ is a translation of the parent graph, right 2 and up 7.

7.



8.



9. Her value of a is reasonable. Since the function was reflected over the x -axis and shrunk vertically it must be true that $-1 < a < 0$. Her value of b is not reasonable because that would cause the function to shift to the right. The value of b must be negative. Her value of c is also reasonable since shifting up requires a positive value of c .

10.

